

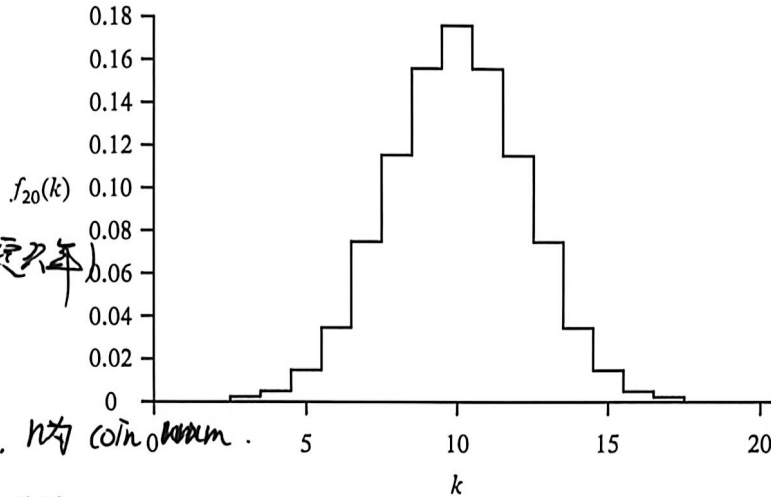


Figure 18.4 The pdf for the unbiased binomial distribution for $n = 20$, $f_{20}(k)$.

prob. 每个硬币 (并得一定年)
承而证明其独立.

⇒ 样本空间 2^n 个结果, n 为 coin num.

$$P = p^k \times q^{(n-k)}, \quad q = (1-p)$$

设 A B 分别为正, 负为反面.

$$Pr(A \cap B) = p \times p \times 1$$

⇒ 设有 1 正面, 剩下 $n-1$

$$Pr = \sum_{k=0}^{n-1} \binom{n-1}{k} \times p^k \times q^{(n-1-k)}$$

$$\Rightarrow (p+q)^{(n-1)}, \quad \text{而 } p+q=1$$

所以剩下来为 $\frac{1}{2}$

$$\text{即 } Pr(A \cap R) = Pr(A) Pr(R)$$

23 heads plus ... the probability of flipping no heads.

The General Binomial Distribution

If the coins are biased so that each coin is heads with probability p , then the number of heads has a *general binomial density function* specified by the pdf $f_{n,p}(k)$: $[0..n] \rightarrow [0, 1]$ where

$$f_{n,p}(k) = \binom{n}{k} p^k (1-p)^{n-k}. \quad \text{硬币 = 硬币} \quad (18.1)$$

for some $n \in \mathbb{N}^+$ and $p \in [0, 1]$. This is because there are $\binom{n}{k}$ sequences with k heads and $n-k$ tails, but now $p^k (1-p)^{n-k}$ is the probability of each such sequence.

For example, the plot in Figure 18.5 shows the probability density function $f_{n,p}(k)$ corresponding to flipping $n = 20$ independent coins that are heads with probability $p = 0.75$. The graph shows that we are most likely to get $k = 15$ heads, as you might expect. Once again, the probability falls off quickly for larger and smaller values of k .

近似估计

Theorem 二项分布近似式

$$Pr(R=k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \text{令 } k = \alpha n,$$

$$\binom{n}{\alpha n} \sim \frac{1}{\sqrt{2\pi n \alpha (1-\alpha)}} \cdot \alpha^{-\alpha n} \cdot (1-\alpha)^{-(1-\alpha)n}$$

$$\Rightarrow Pr(R=\alpha n) \sim \frac{\left[\left(\frac{p}{\alpha}\right)^\alpha \left(\frac{1-p}{1-\alpha}\right)^{1-\alpha} \right]^n}{\sqrt{2\pi n \alpha (1-\alpha)}}, \quad H(\alpha, p) = \alpha \log_2 \left(\frac{\alpha}{p}\right) + (1-\alpha) \log_2 \left(\frac{1-\alpha}{1-p}\right)$$

$$\Rightarrow Pr(R=\alpha n) \sim \frac{1}{\sqrt{2\pi n \alpha (1-\alpha)}} \cdot 2^{-n H(\alpha, p)}$$